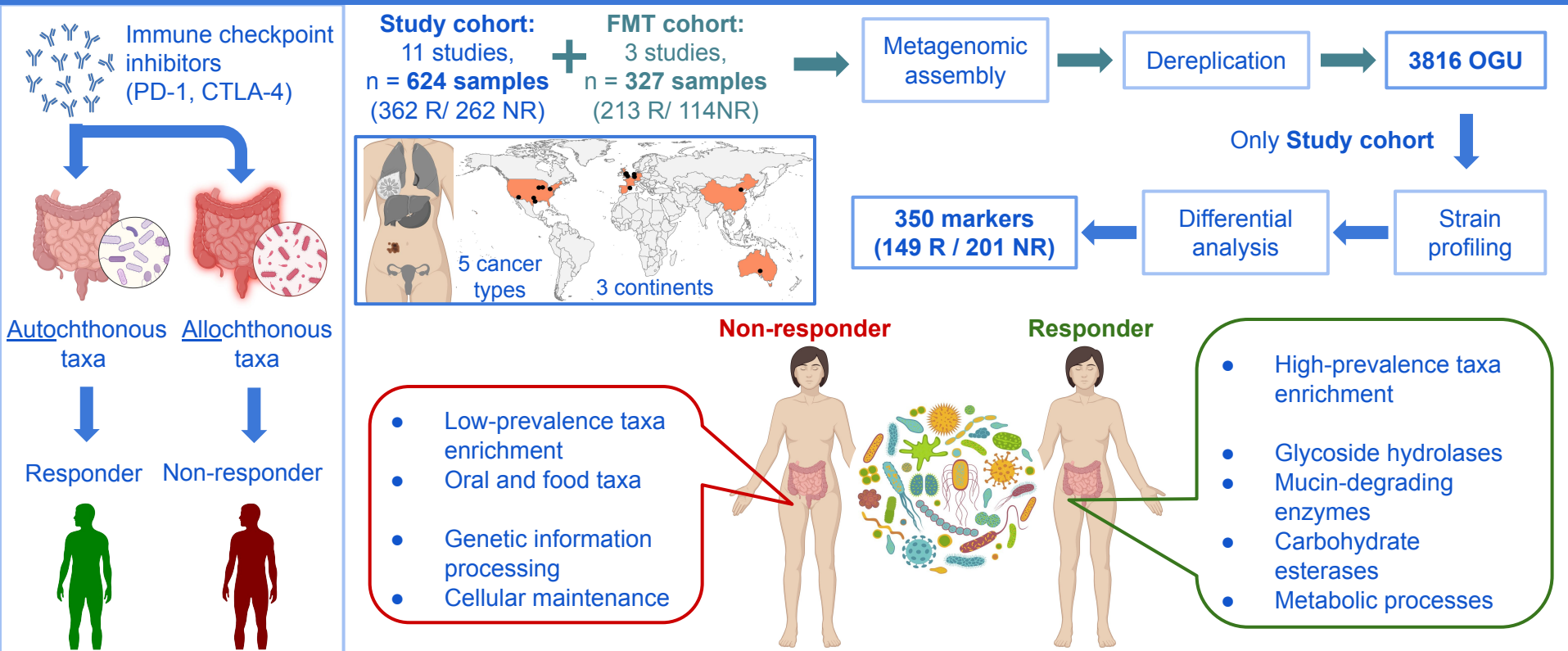


Ecological and functional stratification of the stool microbiome predicts response to immune checkpoint inhibitors across cancer types



Response to immune checkpoint inhibitors is governed by gut ecosystem stability. Clinical benefit associates with a functionally coherent core of prevalent, autochthonous taxa enriched in genes for carbohydrate metabolism and amino acid biosynthesis. Non-response is characterized by variable enrichment of low-prevalence exogenous (oral/food-derived) bacteria and a functional shift toward nucleotide metabolism.